LEVELHULL · STEM CHECK

LevelHull STEM check

Design Molecule Lab · designmolecule.com

DOCUMENT CONTROL

Document ID	LVH-STM-001	Version	0.2
Status	Draft	Date	2026-10-25
Author	Amish Chadha	License	CERN-OHL-S-2.0
Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/levelhull		
Source	docs/01b-stem-check.md @ e0fd582		

REVISION HISTORY

Version	Date	Author	Change
0.2	2026-10-25	Amish Chadha	Amish's decisions of 2026-10-25: S32 within known science with R2 and R3 stated for the free-flooding condition (D-O2); S34 within with four bailers, about 11 min (D-O3); 29 within, 5 plausible, 0 fail
0.1	2026-10-24	Amish Chadha	First STEM check (kit 1.9.1 upgrade) on the TRL 3 design and concept options B and C; two details fixed under Amish's standing instruction (LVH-DEC-001 D-S1, D-S2); independent review recorded; sealed-hull rows S32 and S34 proposed to Amish (D-O2, D-O3)

LevelHull STEM check

Verdict: the working principle of LevelHull (closed-cell foam fixed high inside the hull cuts the waterline of the swamped canoe along both sides, so it floats upright on its own foam) sits within known science, and so do the principles of options B and C. Of the 34 claims, **27 are within known science, 5 are plausible and need field data or a test at TRL 4 and later** (S10 cover and webbing life, S17 sustained bailing rate, S23 decay at screw holes, S27 abrasion of option B's sponsons, S33 roll behaviour of a sealed hull brim-full), and the 2 rows that failed with a fix proposed to Amish (S32, S34) are now within known science: straight after swamping a sealed canoe is brim-full and sits about 24 mm up (about 3 mm at its lowest point), so the 50 mm of R2 and the bail-out time of R4 held only once the water inside was down to the lake level. Amish decided on 2026-10-25 ("i concur with your recommendations") to keep the hull sealed and state R2 and R3 for the free-flooding condition of the USCG method with the operating rule "hold on, then bail at once, seated" (LVH-DEC-001 D-O2), and to supply four bailers so all four crew bail, about 11 min from brim-full (D-O3). Of the 34 claims, 29 are now within known science and 5 plausible. `stem_check` is **open**. Two details were fixed under Amish's standing instruction of 2026-10-23: the end chock screws went only 10 mm into the batten (S13, D-S1), and galvanised coach screws are withdrawn in favour of A4 stainless (S22, D-S2). The independent review also corrected the calculation note's floor comparison and its handling of timber denser than water. Method: STANDARDS section 20. Calculations: `docs/04-calcs/stem_check.py`, which reads LVH-CAL-001 (`docs/04-calcs/sizing.py`, `results.csv`) and the model, and writes `stem_results.csv` and `options_results.csv`.

> **Safety:** LevelHull is flotation for a boat, not lifesaving equipment for people, and it is never a substitute for life jackets. These checks are screening calculations for a paper design; no fitted canoe is used before its fixings are pull-tested and it has passed a swamp test in sheltered, shallow water with a safety boat and every person in a life jacket.

Claims checked

Table 1. Claims, governing science and verdicts (generated from `docs/04-calcs/stem_results.csv` by `stem_check.py`). S1 to S24 and S31 to S34 are option A, the design; S25 to S27 option B; S28 to S30 option C.

#	Claim	Field	Branch	Governing law or source	First-order check	Verdict	Closed by
S1	Swamped canoe at the design load (118 kg net) floats with 113 mm freeboard amidships, 92 mm at the lowest point, with the water inside level with the lake (the USCG swamped condition)	Physics	Fluid mechanics	Archimedes: lift of the foam and timber below the waterline equals the net weight; water inside the flooded hull carries nothing while it flows freely in and out (33 CFR 183.220(g))	claimed 113 mm, estimate 113 mm (ratio 1.00)	Within known science	LVH-CAL-001 [F1], [F2]
S2	Straight after swamping, with the hull brim-full and the water trapped above the lake level, the canoe still floats (24 mm freeboard) and rises as the crew bail	Physics	Fluid mechanics	Reserve at zero freeboard 277 kg over the outer waterplane 11.4 m2 (trapped water rises with the hull, level canoe); a sealed hull drains only over the gunwale, so this is the state until the crew bail (see S32 to S34)	24.3508 mm >= 0 mm	Within known science	This check; LVH-CAL-001 v0.3 section 10
S3	Swamped roll stiffness 761 kg m per radian, metacentric height 3.1 m	Physics	Mechanics and statics	Lost-buoyancy method (water inside flowing freely): K = rho I_T + B z_B - W z_G; hand estimate: rho I_T of the foam strips and planks only. The 3.1 m is K over the net weight carried, not a metacentric height of the whole displacement	claimed 761 kg m/rad, estimate 783 kg m/rad (ratio 0.97)	Within known science	LVH-CAL-001 [F5], [F6]
S4	The same foam laid on the floor gives no roll stiffness with 5 kg more load,	Physics	Mechanics and statics	Foam fully under water leaves only the planks' waterplane; K =	-5 kg m/rad <= 0 kg m/rad	Within known science	LVH-CAL-001 [F17] to [F20]

#	Claim	Field	Branch	Governing law or source	First-order check	Verdict	Closed by
	so the modules must sit high (roll stiffness of the floor arrangement)			rho I_T + B z_B - W z_G			
S5	All four crew holding one grab line heel the swamped canoe 3.9 deg (limit 10)	Physics	Mechanics and statics	Heel = heeling moment / roll stiffness (small angles)	claimed 3.9 deg, estimate 3.93 deg (ratio 0.99)	Within known science	LVH-CAL-001 [F10], [F11]
S6	With the largest module lost and 5 % uptake the canoe still floats (25 mm lowest freeboard)	Physics	Fluid mechanics	Archimedes with one 1,400 mm module removed and foam lift down 5 %	25 mm >= 0 mm	Within known science	LVH-CAL-001 [F14], [F15]
S7	Lowest freeboard stays at or above 50 mm with the design net load 20 % higher and foam lift 5 % lower	Mathematics	Statistics and uncertainty	Re-solve of the LVH-CAL-001 equilibrium with both inputs at their pessimistic values	75.6 mm >= 50 mm	Within known science	This check; LVH-CAL-001 v0.3 section 10
S8	Closed-cell polyethylene foam loses under 5 % of its lift to water uptake after 30 days immersed	Chemistry	Materials and polymer chemistry	Closed-cell PE foam sheet, 33 kg/m3: water absorption under 2 % by volume after 28 days (EN 12087), Ravago Ethafoam data sheet; lift lost = absorbed volume x 1,000 / net lift of 970 kg/m3	2.07 % of lift <= 5 % of lift	Within known science	Cited data sheet; immersion test at TRL 4 (R6)
S9	Polyethylene foam is not dissolved by petrol, diesel or fish slime (polystyrene is excluded for that reason)	Chemistry	Organic	Polymer solubility: polyethylene dissolves only in hot aromatic or chlorinated solvents (above about 80 C); at ambient low-	The foam is about 3 % polymer by volume, so swelling does not change its lift; softening and diesel and fish-slime exposure are covered by	Within known science	LVH-DDR-001 D2

#	Claim	Field	Branch	Governing law or source	First-order check	Verdict	Closed by
				density PE swells and softens a little in petrol; polystyrene dissolves	the cover and the R6 immersion test at TRL 4		
S10	PVC-coated tarpaulin covers and polyester webbing keep their strength for 12 months in tropical sun and water	Chemistry	Materials and polymer chemistry	UV photo-degradation of PVC coatings and polyester fibres; UV-stabilised grades are sold for outdoor use, but strength retention after a year in equatorial sun depends on the grade	Cannot be shown by calculation; strap and cover inspection in field use (R9, TRL 4 and later)	Plausible, to confirm at TRL 3	Open: needs field data (TRL 4 and later)
S11	Each M8 coach screw carries twice its share of the full lift (0.30 kN): at twice the load 0.60 kN, of which 0.51 kN along the frame face and 0.32 kN pulling the batten off it (static; wave slam not included)	Engineering and technology	Standards practice	Interaction (F_ax / R_ax)^2 + (F_v / R_v)^2 <= 1 at twice the load; R_v 1.5 kN (LVH-CAL-001 A9, conservative against a Johansen estimate of about 5 kN), R_ax 8.0 kN from the EN 1995-1-1 withdrawal expression for screws, 600 kg/m3 hardwood, 48 mm penetration	0.118 <= 1	Within known science	LVH-CAL-001 [S5], [S6]; pull test at TRL 4 (R7)
S12	Webbing and D-ring buckles carry twice the full lift of the largest module (lowest factor, D-rings)	Physics	Strength of materials	Load per strap = full lift / 2; parts specified at 10 kN webbing and 4 kN D-ring pair (BOM lines 8, 9, to confirm with real parts)	13.3 >= 2	Within known science	LVH-CAL-001 [S3], [S4]
S13	End chock screws (5 x 60 mm) bite far enough into the	Engineering and technology	Standards practice	Pointside penetration of at least 6 d (30 mm for a 5 mm	31.3 mm >= 30 mm	Within known science	LVH-DEC-001 D-S1 (was Fails, fix

#	Claim	Field	Branch	Governing law or source	First-order check	Verdict	Closed by
	batten: 31 mm (was 10 mm when driven straight down through the 50 mm chock)			screw), EN 1995-1-1 practice for laterally loaded nails and screws; screws now driven parallel to the frame face in 12 mm counterbores 30 mm deep; margin 1.3 mm, so the counterbore depth is a making check (LVH-DWG-103)			proposed: 10 mm)
S14	Coach screws go 48 mm into the frames and stop 12 mm short of the planking (no new hole below the waterline)	Engineering and technology	Manufacturing tolerances	Penetration at least 6 d = 48 mm (EN 1995-1-1 practice); frame moulded depth 60 mm; planking untouched; no margin either way, so each frame's depth is measured and the screw length chosen to suit (LVH-DEC-001, to confirm item 2)	48 mm >= 48 mm	Within known science	Model check (cad/src/model.py)
S15	Grab-line eye bolt (M8 x 100) passes through strake, plank, two washers and a nyloc nut with thread to spare (2.9 mm)	Engineering and technology	Manufacturing tolerances	Grip at 625 mm: strake and top plank; ISO 7089 washers 1.6 mm, ISO 7040 nut 9.2 mm; at least one 1.25 mm pitch beyond the nut	2.9 mm >= 1.25 mm	Within known science	Model check (cad/src/model.py)
S16	A crew member holding on with head and shoulders out puts about 15 kg (20 % of 75 kg) on the canoe	Biology	Human physiology and biomechanics	Body segment masses (Winter): head and neck 8.1 %, forearms and hands on the gunwale about 5 %; the body below the water is close to neutral	claimed 15 kg, estimate 9.83 kg (ratio 1.53)	Within known science	LVH-CAL-001 A3

#	Claim	Field	Branch	Governing law or source	First-order check	Verdict	Closed by
S17	4 crew bail at 100 L per minute each (1.17 m3 in about 3 min)	Biology	Ergonomics	Lifting work is small (about 11 W against 75 W sustained); the limit is the cycle: 100 L/min is about 14 scoops a minute for many minutes; sustained rates of 30 to 50 L/min are more usual (independent review)	Rate to measure in the bail-out trial (R4, TRL 4); see S34	Plausible, to confirm at TRL 3	Open: needs field data (TRL 4 and later)
S18	Partly bailed to 200 mm freeboard with all crew aboard and 2.3 m3 of water still inside, the canoe keeps a positive metacentric height (0.14 m after the free-surface loss)	Physics	Mechanics and statics	GM = KB + I_T / V - KG - i_fs / V: KB 0.25 m, waterplane I_T 1.38 m4, free surface i_fs 0.98 m4 of a 1.19 m wide pool, KG 0.22 m with crew seated (CG 0.68 m)	0.142 m >= 0 m	Within known science	This check; LVH-CAL-001 v0.3 section 10
S19	The only new hull holes (eye bolts, 625 mm up) stay above the loaded waterline of the dry canoe (175 mm)	Engineering and technology	Standards practice	Dry displacement 1068 kg (timber at 750 kg/m3, crew, motor, fuel, gear, catch, kit) on the model hull	449.937 mm above >= 0 mm above	Within known science	Model (cad/src/model.py)
S20	Modules take 8.4 % of the inside volume below the gunwale	Mathematics	Units and dimensional analysis	0.421 m3 of module envelope over 5.01 m3 inside volume	8.4 % <= 10 %	Within known science	LVH-CAL-001 [V2]
S21	Kit mass 70 kg per canoe	Mathematics	Orders of magnitude	Sum of stock quantities in the BOM times densities	claimed 69.9 kg, estimate 69.5 kg (ratio 1.01)	Within known science	LVH-CAL-001 [K1]
S22	Stainless kit hardware does	Chemistry	Electrochemistry and batteries	All metal in contact is A4	0 dissimilar couples <= 0		BOM lines 3, 10; LVH-

#	Claim	Field	Branch	Governing law or source	First-order check	Verdict	Closed by
	not corrode the fixings away in wet hardwood in fresh or salt water			stainless (screws, D-rings, eye bolts): no galvanic couple. A4 can suffer crevice corrosion in wet, salty, airless timber, so screws are inspected at each refit; zinc coatings are attacked by tannins in fresh water too, so the BOM's hot-dip galvanised alternative is withdrawn in favour of A4 (LVH-DEC-001 D-S2)	dissimilar couples	Within known science	DEC-001 D-S2
S23	Fungal decay does not start at the new screw holes in the frames within the kit's life	Biology	Microbiology	Wood-decay fungi need timber moisture above about 20 %; frames of a working canoe are wet, so the holes are sealed with preservative or epoxy (BOM line 14) and inspected	Depends on timber and care; inspection at each refit	Plausible, to confirm at TRL 3	Open: needs field data (TRL 4 and later)
S24	A builder and a helper fit the kit in about 6.7 h	Engineering and technology	Manufacturing tolerances	Bottom-up count by the author (not independent): 20 marks, 8 battens, 20 screws, 16 counterbored chocks, 16 straps, 8 eye bolts, marks; sealer cure not included	claimed 6.7 h, estimate 6.73 h (ratio 1.00)	Within known science	LVH-CAL-001 [T1] (estimate, R8)
S31	Lowest freeboard stays at or above 50 mm if the wet hull timber is denser than water, 1,100	Physics	Fluid mechanics	Saturated hardwood can exceed 1,000 kg/m3, so A2 is neutral, not conservative; re-solve at 1,100 and	75 mm >= 50 mm	Within known science	This check

#	Claim	Field	Branch	Governing law or source	First-order check	Verdict	Closed by
	kg/m3 (75 mm; 57 mm at 1,200 kg/m3)			1,200 kg/m3; the real hull is weighed before fitting (LVH-DEC-001, to confirm item 3)			
S32	R2 as restated for the free-flooding condition: the lowest point keeps 50 mm with the water inside at the lake level; brim-full, a sealed hull still floats and the crew bail at once, seated	Physics	Fluid mechanics	Brim-full freeboard (S2) less the trim drop to the lowest point of LVH-CAL-001 (21 mm)	24 mm level, about 3 mm at the lowest point brim-full; R2 is met once the water inside is down to the lake level (by bailing or by water spilling out as the canoe rolls). Amish decided on 2026-10-25 to keep the hull sealed and state R2 and R3 for the free-flooding condition of the USCG method, with the operating rule "hold on, then bail at once, seated"; freeing ports studied at TRL 4 (LVH-DEC-001 D-O2)	Within known science	LVH-DEC-001 D-O2
S33	The swamped canoe floats level (R3) whether or not the water inside can flow in and out	Physics	Mechanics and statics	Sealed hull, water trapped: K = rho (I_out - i_fs) + B z_B - W z_G; brim-full: I_out 2.14 m4, free surface 2.01 m4; at the lake level: I_out 1.73 m4, free surface 0.46 m4 between the modules	Roll stiffness about 70 kg m/rad brim-full (near neutral; the estimate of the centre of weight is good to about +-200) and 1213 kg m/rad once the water inside is down to the lake level, where the modules cut its free surface. Brim-full, the canoe lolls a	Plausible, to confirm at TRL 3	Open: needs field data (TRL 4 and later)

#	Claim	Field	Branch	Governing law or source	First-order check	Verdict	Closed by
					degree or two until the low gunwale dips and water spills out, which lowers the inside level and stiffens it; to confirm in the swamp test of a sealed hull (R3, TRL 4)		
S34	The four crew, each with a bailer, bail the brim-full canoe to 200 mm freeboard within 15 min (R4) at a sustained 50 L/ min each	Biology	Ergonomics	Brim-full water 4.47 m3 less the 2.31 m3 that may stay aboard	2.16 m3 at 4 x 50 L/min takes about 11 min (two bailers took about 22 min). Four bailers decided by Amish on 2026-10-25 (LVH-DEC-001 D-O3); rate to measure in the bail-out trial	Within known science	LVH-DEC-001 D-O3
S25	Option B: swamped canoe on external sponsons floats with 158 mm lowest freeboard	Physics	Fluid mechanics	Same equilibrium as LVH-CAL-001 with the foam outside the planks, 360 to 560 mm up	158 mm >= 50 mm	Within known science	This check
S26	Option B: the new bolt holes through the planks (325 mm up) and the sponson bottoms (360 mm) stay above the loaded dry waterline	Engineering and technology	Standards practice	Dry loaded draft on the model hull, as S19. The holes are below the swamped waterline, so they must be bedded watertight or they leak while the canoe is bailed	150 mm above >= 0 mm above	Within known science	This check
S27	Option B: sponsons survive beaching, rafting alongside other	Engineering and technology	Standards practice	External fenders and sponsons on working boats wear and tear at the chine and ends; the	Abrasion life to establish in field use	Plausible, to confirm at TRL 3	Open: needs field data (TRL 4 and later)

#	Claim	Field	Branch	Governing law or source	First-order check	Verdict	Closed by
	canoes and wave slap			tarpaulin cover is the only skin			
S28	Option C: swamped canoe hung between two pipe floats has 160 mm lowest freeboard	Physics	Fluid mechanics	Same equilibrium with two 250 mm floats 2.1 m off the centreline; booms and struts counted as weight	160 mm >= 50 mm	Within known science	This check
S29	Option C: each boom carries half the full lift of one float pushed under by a wave (2.15 kN) at 1280 mm from the gunwale bolts: 8.5 MPa	Physics	Strength of materials	Cantilever, sigma = M / Z, Z = b d^2 / 6 (the boom is continuous across the canoe, so the gunwale bolts take the uplift of the couple, about M / 1.64 m); hardwood of strength class D30, 30 MPa characteristic, design value about 30 x 0.7 / 1.3 = 16 MPa (EN 338, EN 1995-1-1 modification and material factors)	8.5 MPa <= 16 MPa	Within known science	This check
S30	Option C: the SDR 26 float pipe does not collapse when pushed 0.5 m under (3.0 m of water collapse head)	Physics	Strength of materials	Ring buckling of an unsupported pipe p_c = 2 E / (1 - nu^2) (t / (D - t))^3, long-term E of PE100 about 200 MPa, nu 0.4	3.03 m of water >= 0.5 m of water	Within known science	This check

Verdicts: **Within known science**, **Plausible, to confirm at TRL 3**, **Fails, fix proposed**, **Fails**.

Fixes made under the standing instruction

Table 2. Fixes (LVH-DEC-001).

Item	What failed	Fix	Result
End chock screws (S13)	Two 5 x 60 mm wood screws driven straight down through a 50 mm chock leave only 10 mm in the batten, well short of the 6 d (30 mm) penetration of timber practice. Straight down, a deeper screw would also break out of the batten's sloping inboard face within 30 mm	D-S1: drive both screws down parallel to the frame face (31.6 deg off vertical on the reference canoe), 15 and 35 mm out from it, in 12 mm counterbores 30 mm deep. Same screws, same chock, no cost change	31 mm in the batten; the screw stays inside the chock and batten for its full length (model, BOM line 4, build plan step for the chocks, LVH-DWG-103)
Coach screw material (S22)	The BOM allowed hot-dip galvanised coach screws in fresh water; tannins in wet hardwood attack zinc in fresh water too (independent review)	D-S2: A4 stainless only, inspected for crevice corrosion at each refit; no cost change	BOM line 3, build plan section 3

Neither fix changes the working principle, the pitch or the scope. Both are recorded in LVH-DEC-001 as decided under Amish's standing instruction of 2026-10-23, and he can reverse them.

Fixes proposed to Amish

Row	What fails	Options (LVH-DEC-001)
S32	A sealed canoe filled to the brim keeps about 24 mm of freeboard level and about 3 mm at its lowest point until the crew bail; R2's 50 mm holds only once the water inside is down to the lake level (as in the USCG test, where water flows freely in and out)	D-O2: (a) one-way freeing ports in the topsides above the dry waterline; (b) restate R2 and R3 for the free-flooding condition, with the operating rule "hold on, then bail at once, seated"; (c) more foam, which helps little. Decided 2026-10-25: (b), with (a) studied at TRL 4
S34	From brim-full, 2.1 m3 must be bailed to reach 200 mm freeboard; at a sustained 50 L/min each, two crew take about 22 min against R4's 15	D-O3: (a) four bailers, all four crew bail (about 11 min); (b) restate R4; (c) a hand bilge pump. Decided 2026-10-25: (a)

Results added by this check

- **Straight after swamping (S2, S32, S33).** The calculation note, like the USCG level flotation method (33 CFR 183.220(g): "allowing water to flow between the inside and outside of the boat"), treats the water inside as part of the lake. A planked canoe has no opening below the gunwale, so after a wave fills it to the brim the water inside stays above the lake level and rises with the hull. The reserve at zero freeboard (277 kg) over the outer waterplane (about 11.4 m2) then holds the gunwale about 24 mm up, about 3 mm at the low end. In that state the canoe is near neutral in roll (S33), so it lolls a degree or two until a gunwale dips and water spills out, which lowers the inside level; bailing does the rest, and at the lake level the 92 mm of LVH-CAL-001 follows. The crew should expect a canoe nearly awash at first.
- **Pessimistic inputs (S7, S31).** Design net load 20 % higher and foam lift 5 % lower together: 76 mm at the lowest point. Wet timber at 1,100 or 1,200 kg/m3 instead of 1,000: 75 and 57 mm. All above the 50 mm of R2 in the lake-level condition.
- **Partly bailed (S18).** At 200 mm freeboard with all four crew aboard and 2.3 m3 of water still inside, the free surface of that water costs about 0.28 m of metacentric height; 0.14 m remains, with the crew seated (the reviewer's centre of weight gives about 0.10 m). Crews bail sitting low, not standing.

Fields not relevant

None. Every field is touched: physics (hydrostatics, stability, strength of fixings and booms), chemistry (foam water uptake and fuel resistance, polymer UV life, corrosion), biology (crew physiology, bailing effort, wood decay), mathematics (units, orders of magnitude, sensitivity) and engineering (fastener practice, tolerances, fitting time). Branches such as electromagnetism, optics, acoustics and nuclear radiation are not relevant: the kit has no electrical, optical or radiating parts.

Concept options

- **Option A, foam modules high inside (the design):** S1 to S24 and S31 to S34. Within known science except the field items (S10, S17, S23, S33) ; the two sealed-hull rows S32 and S34 are within known science since Amish's decisions D-O2 and D-O3 of 2026-10-25.
- **Option B, external foam sponsons:** within known science (S25, S26): 158 mm lowest freeboard and twice A's roll stiffness, because the same foam sits lower and further out. Its 32 bolt holes through the planks are below the swamped waterline and must be bedded watertight. S27, abrasion and tearing of the external covers, is plausible, to confirm in field use. Losing one of its four sponsons sinks the gunwale 32 mm under (R12 not met as drawn); see LVH-OPT-001.
- **Option C, double outrigger floats:** within known science (S28 to S30): 160 mm lowest freeboard, very high roll stiffness, booms at 8.5 MPa against 16 MPa when a float is driven under, floats safe against collapse. Losing one float lets the canoe capsize; see LVH-OPT-001.

The brim-full condition of S2 and S32 to S34 applies to all three options: B and C also have sealed hulls. No option has a Fails row; none is dropped on science.

Independent review

A second agent (a separate headless Claude session, `claude -p`, with no tools) was given only four files, this STEM check, the calculation note LVH-CAL-001, the product requirements LVH-PRS-001 and `sizing.py`, and none of the author's reasoning, and asked on 2026-10-24 to refute the headline results and every "Within known science" verdict. It recomputed most of them by hand. What it found and what changed:

Item	Finding	Change
Headline 113 and 92 mm (S1)	Arithmetic upheld (113.3 mm by hand). Weakened: a planked canoe is sealed below the gunwale, so the water inside does not fall to the lake level by itself; the real state after swamping is brim-full	S2 restated; new S32 (R2 brim-full, fails, to Amish D-O2) and S33 (roll, plausible); R2 and R3 statuses changed in LVH-PRS-001
S2 "evens out in any sea"	Refuted: water above the lake level leaves only over the gunwale	Wording removed; S33 shows the brim-full canoe is near neutral in roll and spills over the low gunwale as it lolls
Wet timber at 1,000 kg/m3 (A2)	Weakened: saturated hardwood can exceed 1,000 kg/m3, so A2 is neutral, not conservative	New S31: 75 mm at 1,100 and 57 mm at 1,200 kg/m3; the script's trim now carries the weight moment of denser timber (it did not); hull weighing stays an item to confirm
Lift on the envelope, not the core	Weakened: if the sleeves flood, lift is about 4.5 % less	Bounded by the 5 % case [F13] (86 mm); stated in LVH-CAL-001 section 10
Floor comparison (S4, F17 to F20)	Weakened: the script put the floor case's waterplane at the high position; the floor arrangement floats higher (about 360 mm)	Script corrected: 135 kg m per radian and 22 deg with the crew on one side (was 283 and 10.6 deg); trade stated as freeboard against stiffness; the case for high modules is stronger
Bailing (S17, R4)	Weakened: from brim-full there is 2.1 m3 to bail, and 100 L/min each is optimistic; 30 to 50 L/min is more usual	S17 now plausible (rate to measure); new S34: about 22 min at 50 L/min each with two bailers, failed R4; four bailers decided by Amish on 2026-10-25 (D-O3), about 11 min
S3 roll stiffness	Upheld; the hand check left out the planks	Hand check includes the planks; the "metacentric height" label explained
S11 fixings	Upheld; the cell text gave the undoubled loads	Text corrected to the doubled loads
S9 fuel	Upheld; reasoning wrong (the foam is about 3 % polymer, so swelling does not change lift)	Reasoning corrected
S13, S14	Weakened: 1.3 mm and zero margin on the penetration rules	Margins stated; counterbore depth and frame depth become making and measuring checks
S18 partly bailed	Weakened: its centre of weight gives about 0.10 m, and intermediate states are not scanned	Kept within (positive either way); crews bail seated (hazard H6)
S22 corrosion	Weakened: the risk is crevice corrosion of A4, and the BOM's galvanised alternative contradicts the row	D-S2: galvanised withdrawn; inspection at each refit
S24 fitting time	Weakened: not independent; the chock counterbores were not counted	Chocks 60 min: 6.7 h; marked as the author's estimate
S26 option B	Weakened: the bolt holes are below the swamped waterline	Noted in S26 and LVH-OPT-001
S29 option C	Weakened: the gunwale bolts take the boom's couple	Noted: the boom is continuous, so the uplift on the bolts is about M / 1.64 m
	Upheld	None

Item	Finding	Change
S5, S6, S8, S12, S15, S16, S19 to S21, S30		
Not checked	Dynamic (slam) loads on the fixings; frame-to-plank fastenings; foam creep under strap tension; dry stability with the kit	Carried as open items in `docs/REVIEW.md` and the design FMEA

The reviewer would also have the 27 "within" rows that rest on the author's own script reclassified as plausible. They stay within where an independent hand check agrees (S1, S3, S5, S21) or the margin is clear; the rows it refuted or weakened on substance are changed as above.

Sources

- US Coast Guard, *Safe Boating: Flotation* guideline and 33 CFR 183.220 to 183.235 (level flotation test, swamped condition), https://www.uscgboating.org/assets/PDF/downloads/FLOTATION.pdf, accessed 2026-10-25.
- Ravago Building Solutions, "Ethafoam technical data sheet" (density 33 kg/m3, ISO 845; water absorption under 2 % by volume after 28 days, EN 12087), https://ravagobuildingsolutions.com/gr/wp-content/uploads/sites/47/2022/01/tds-ethafoam-eng.pdf, accessed 2026-10-25.
- CEN, EN 1995-1-1, *Eurocode 5: Design of timber structures, Part 1-1: General, common rules and rules for buildings* (withdrawal capacity of screws, penetration rules, modification and material factors).
- CEN, EN 338, *Structural timber: Strength classes* (D30 hardwood, 30 MPa bending, characteristic).
- D. A. Winter, *Biomechanics and Motor Control of Human Movement*, 4th ed. (Hoboken: Wiley, 2009), body segment parameters.
- LevelHull calculation note LVH-CAL-001 (`docs/04-calcs/01-sizing.md`) for every option A figure re-checked here.